

Computer Science & Information Technology

Computer Networks

DPP: 1

Transport Layer

- Q1** UDP provides _____ services.
(A) 'Connection Less' and 'Unreliable'
(B) 'Connection Less' and 'Reliable'
(C) 'Connection Oriented' and 'Unreliable'
(D) 'Connection Oriented' and 'Reliable'
- Q2** Consider total length field value of UDP packet header is 200 (in decimal) calculate its payload size (in bytes)?
- Q3** Which transport layer protocol is used for faster communication?
(A) TCP
(B) UDP
(C) IP
(D) None of the above
- Q4** Which transport layer protocol is used by HTTP/1.1 (version 1.1)?
(A) TCP
(B) UDP
(C) IP
(D) None of the above
- Q5** Identify INCORRECT statement(s) regarding TCP and UDP.
(A) Flow control provided by both TCP and UDP
(B) Flow control provided by TCP, but not UDP
(C) Flow control provided by UDP, but not TCP
(D) Flow control provided by neither TCP nor UDP
- Q6** Consider the 3-way handshake protocol for TCP connection establishment between hosts S and D. Host S sends a TCP connection request message (TCP segment) to D, and D accepts the connection request and send accept message (TCP segment) to S. Identify correct TCP header flag bits in accept message (TCP segment)?
(A) SYN bit = 0 and ACK bit = 0
(B) SYN bit = 0 and ACK bit = 1
(C) SYN bit = 1 and ACK bit = 0
(D) SYN bit = 1 and ACK bit = 1
- Q7** Consider the 3-way handshake protocol for TCP connection establishment between hosts S and D. Host S sends a TCP connection request message (TCP segment) to D. D accepts the connection request and send accept message (TCP segment) to S and waiting for ACK of it from S. Identify the current TCP state of Host D?
(A) LISTEN (B) SYN_SENT
(C) SYN_RECV (D) ESTABLISHED
- Q8** Consider a TCP sender sends one segment in which sequence number field value (in decimal) is 2500 and the segment carried 700 bytes of data in its payload. What should be the acknowledgment number sent by receiver to acknowledge this segment successfully?
- Q9** Consider a TCP sender send two segments to TCP receiver one after another. The sequence number field value in the header of the segments (in decimal) are 7500 and 8800 respectively. How many bytes carried by the first segment in its payload field?



- Q10** Consider TCP protocol runs over a 300 Mbps network. What should be maximum segment lifetime (in seconds, rounded off to nearest integer)?
- Q11** Consider new reliable byte-stream transport protocol like TCP, runs over a 200 Mbps network and maximum segment lifetime is 600 seconds. Calculate minimum number of bits required for sequence number field in the header?
- Q12** Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. The TCP server send ACK to acknowledges the FIN segment, in which state does the server side TCP connection wait to close the connection?
- (A) CLOSE-WAIT (B) TIME-WAIT
(C) FIN-WAIT-1 (D) FIN-WAIT-2
- Q13** Consider size of congestion window of a TCP connection be 16 MSS when a timeout occurs.

The round trip time of the connection is 3 seconds. The time taken (in sec) by the TCP connection to reach up to 12 MSS congestion window is _____.

- Q14** Consider in TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm, where the window size at the start of the slow start phase is 1 MSS and the threshold at the start of the first transmission is 32 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size (in MSS) at the end of the tenth transmission?
- Q15** Consider 'window scale' option is opted during TCP connection establishment phase, the maximum possible window size with window scale option is $(2^{16} - 1) * 2^N$. What is the value of N?
- Q16** Which of the following socket API function is/are executed by TCP server?
- (A) bind() (B) connect()
(C) listen() (D) accept()



Answer Key

Q1 (A)
Q2 192~192
Q3 (B)
Q4 (A)
Q5 (A, C, D)
Q6 (D)
Q7 (C)
Q8 3200~3200

Q9 1300~1300
Q10 114~115
Q11 34~34
Q12 (A)
Q13 21~24
Q14 10~10
Q15 14~14
Q16 (A, C, D)



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Hints & Solutions

Q1 Text Solution:

Connection Less' and 'Unreliable'

Q2 Text Solution:

UDP Segment size = Total length = 200 bytes

UDP Header size = 8 byte

UDP Segment Payload size = (200 – 8) bytes = 192 bytes

Q3 Text Solution:

TCP = UDP + Extra Services

UDP = Basic Protocol

IP = Network Protocol

Q4 Text Solution:

TCP

Q5 Text Solution:

Flow control provided by both TCP and UDP →

False

Flow control provided by TCP, but not UDP →

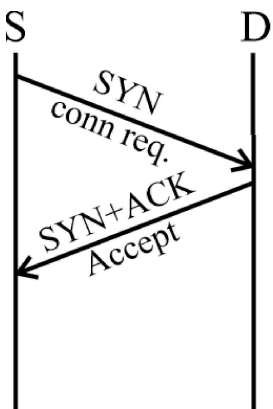
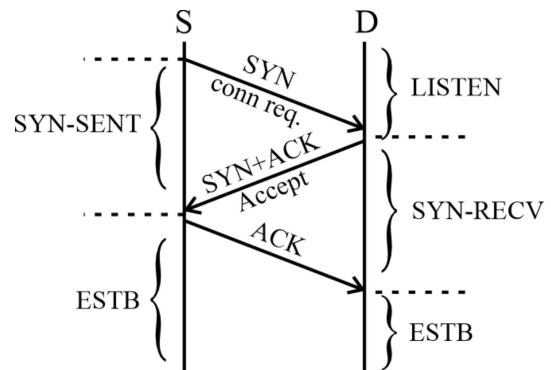
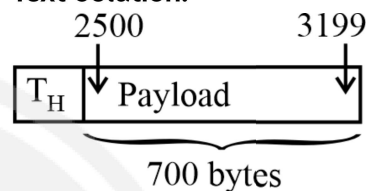
True

Flow control provided by UDP, but not TCP →

False

Flow control provided by neither TCP nor UDP →

False

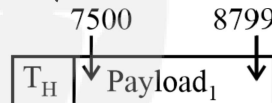
Q6 Text Solution:

Q7 Text Solution:

Q8 Text Solution:


Seq. no. = 2500

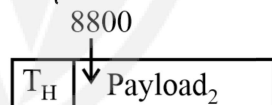
Ack No. = 2500 + 700 = 3200

Q9 Text Solution:

Seq. no. = 7500



Seq. no. = 8800



Payload₁ size = (8800 – 7500) bytes = 1300 bytes

Q10 Text Solution:

$$MSL = \frac{2^{32} \text{ bytes}}{\text{Bandwidth}} = \frac{2^{32} \times 2^3 \text{ bits}}{300 \text{ Mbps}} = \frac{2^{35} \text{ bits}}{3 \times 10^8 \text{ bits/sec}} = 114.53 \text{ sec}$$

Q11 Text Solution:

Maximum no. of bytes can be transmitted in one MSL

= MSL × Bandwidth = 600 sec × 200 Mbps

= $12 \times 10^{10} \text{ bits} = \frac{12 \times 10^{10}}{8} \text{ bytes}$

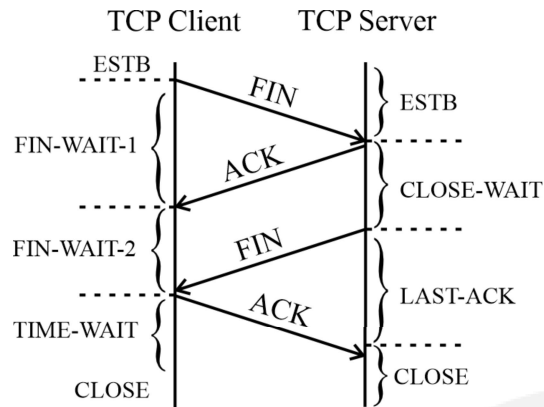


Minimum no. of bits for seq. no. field

$$= \left\lceil \log_2 \left(\frac{12 \times 10^{10}}{8} \right) \right\rceil \text{ bits} = \lceil 33.80 \rceil \text{ bits}$$

$$= 34 \text{ bits}$$

Q12 Text Solution:



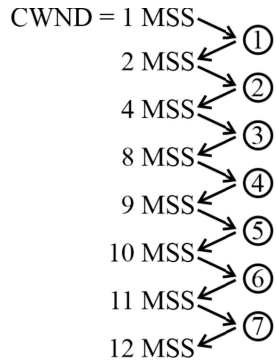
Q13 Text Solution:

CWND = 16 MSS

Time-out ← *

$$SSThresh = \frac{CWND}{2} = \frac{16 \text{ MSS}}{2} = 8 \text{ MSS}$$

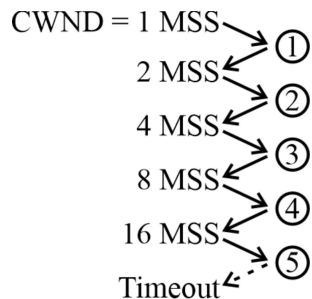
RTT = 3 sec



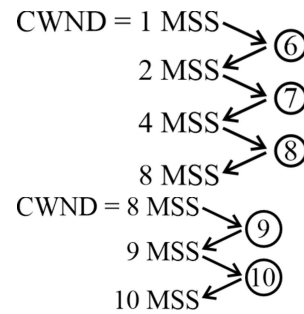
$$\text{Time} = 7 \times \text{RTT} = 7 \times 3 \text{ sec} = 21 \text{ sec}$$

Q14 Text Solution:

SSThresh = 32 MSS



$$SSThresh = \frac{CWND}{2} = \frac{16 \text{ MSS}}{2} = 8 \text{ MSS}$$



Q15 Text Solution:

Maximum possible window(with window scaling) = $(2^{16} - 1) \times \text{Scale Factor}$

Scale Factor = $2^{\text{Shift_Count}}$

In TCP maximum value for shift count can be 14.

Q16 Text Solution:

connect() → TCP client

